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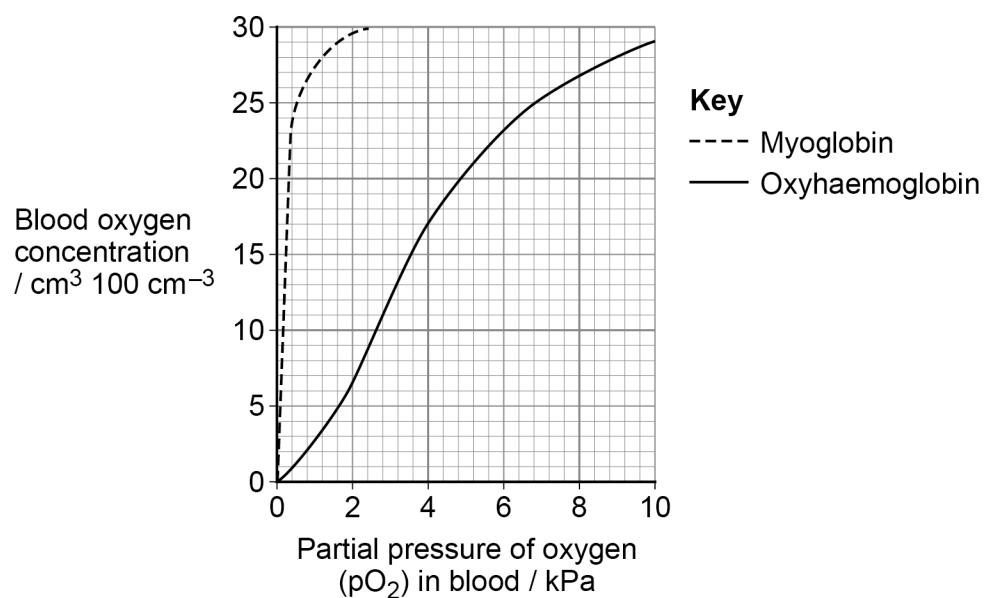
Describe and explain the effect of increasing carbon dioxide concentration on the dissociation of oxyhaemoglobin.

[2 marks]

Seals are diving mammals. They fill their lungs with air before they dive and hold their breath during the dive.

Figure 3 shows the dissociation curves for seal oxyhaemoglobin and seal myoglobin. Myoglobin is an oxygen-carrying protein found in muscles.

Figure 3



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Use information in **Figure 3** to explain how the seal's myoglobin dissociation curve shows the seal is adapted for diving.

[2 marks]

[Extra space]

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Scientists measured the oxygen carrying capacity of seal blood. They found the haemoglobin in a 190 kg seal contained $1.07 \times 10^4 \text{ cm}^3$ oxygen. When the seal dived, it used 5.2 cm^3 oxygen per minute per kg of body mass.

Use this information to calculate the maximum number of minutes the seal can remain under water. Assume that all of the oxygen attached to the haemoglobin is released during the dive.

[2 marks]

Answer = _____ minutes

6

Turn over ►

